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Semiconductor package with substrate recess and methods for forming the same

Abstract

Semiconductor packages and methods of fabricating semiconductor packages include a package substrate having a recess formed in a surface of the package substrate and at least one channel in a bottom surface of the recess. The recess may be configured to accommodate a semiconductor device located over a surface of an interposer that is bonded to the package substrate. Accordingly, a minimum gap distance may be maintained between the semiconductor device and the package substrate, which may ensure that sufficient underfill material may flow between the semiconductor device and the package substrate and within the at least one channel, thereby improving of the structural coupling between the interposer and the package substrate, and reducing the likelihood of package defects, such as delamination, cracking, and/or popcorn defects.

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Background/Summary

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications. Some examples may include personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductive layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon. Dozens or hundreds of integrated circuits are typically manufactured on a single semiconductor wafer, and individual dies on the wafer are singulated by sawing between the integrated circuits along a scribe line. The individual dies are typically packaged separately, for example, in multi-chip modules, or in other types of packaging.

(2) As semiconductor packages have become more complex, ensuring mechanical integrity of the package has become more difficult. Package defects, such as delamination, cracking and other defects, may negatively affect package yields and device performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a vertical cross-section view of an exemplary intermediate structure during a process of forming a semiconductor package including an organic interposer located over a first carrier substrate according to various embodiments of the present disclosure.
- (3) FIG. 2 is a vertical cross-section view of the exemplary intermediate structure showing bonding structures located over the first side surface of the interposer according to various embodiments of the present disclosure.
- (4) FIG. 3 is a vertical cross-section view of the exemplary intermediate structure showing a plurality of semiconductor integrated circuit (IC) dies mounted over the first side surface of the interposer according to various embodiments of the present disclosure.
- (5) FIG. 4 is a vertical cross-section view of the exemplary intermediate structure showing a first underfill material portion located between the lower surfaces of the semiconductor IC dies and the first side surface of the interposer, and a molding portion around the outer periphery of the plurality of semiconductor IC dies according to various embodiments of the present disclosure.
- (6) FIG. 5 is a vertical cross-section view of the exemplary intermediate structure showing a second release layer located over the upper surfaces of the plurality of semiconductor dies, the exposed upper surface of the first underfill material portion and the exposed upper surface of the molding portion, and a second carrier substrate over the second release layer according to various embodiments of the present disclosure.
- (7) FIG. 6 is a vertical cross-section view of the exemplary intermediate structure showing the first carrier substrate removed according to various embodiments of the present disclosure.
- (8) FIG. 7A is a vertical cross-section view of the exemplary intermediate structure showing a plurality of bonding pads and semiconductor devices located over the second side surface of the interposer according to various embodiments of the present disclosure.
- (9) FIG. 7B is a top view of the exemplary intermediate structure of FIG. 7A.
- (10) FIG. 8A is a vertical cross-section view of a package substrate according to various embodiments of the present disclosure.
- (11) FIG. 8B is a top view of the package substrate of FIG. 8A.
- (12) FIG. 9A is a vertical cross-section view of the package substrate showing a plurality of recesses formed in the first side surface of the package substrate according to various embodiments of the present disclosure.
- (13) FIG. 9B is a top view of the package substrate of FIG. 9A.
- (14) FIG. 9C is a top view of an alternative embodiment package substrate of FIG. 9A.
- (15) FIG. 9D is a top view of another alternative embodiment of the package substrate of FIG. 9A.
- (16) FIG. 10A is a vertical cross-section view of a package substrate including a plurality of recesses formed in the first side surface of the package substrate extending partially into a redistribution layer of the package substrate according to an embodiment of the present disclosure.
- (17) FIG. 10B is a vertical cross-section view of a package substrate including a plurality of recesses formed in the first side surface of the package substrate extending partially into a substrate core of the package substrate according to an embodiment of the present disclosure.
- (18) FIG. 11A is a vertical cross-section view of the package substrate showing a channel formed in the bottom surface of each of the recesses according to an embodiment of the present disclosure.
- (19) FIG. 11B is a top view of the package substrate of FIG. 11A.
- (20) FIG. 11C is a top view of an alternative embodiment package substrate of FIG. 11A.
- (21) FIG. 11D is a top view of another alternative embodiment of the package substrate of FIG.

11A.

(22) FIG. 11E is a top view of another alternative embodiment of the package substrate of FIG. 11A.

(23) FIG. 11F is a top view of another alternative embodiment of the package substrate of FIG. 11A.

(24) FIG. 11G is a top view of another alternative embodiment of the package substrate of FIG. 11A.

(25) FIG. 11H is a top view of another alternative embodiment of the package substrate of FIG. 11A.

(26) FIG. 12 is a vertical cross-section view of an exemplary intermediate structure during a process of forming a semiconductor package showing a package structure mounted over the front side surface of a package substrate according to various embodiments of the present disclosure.

(27) FIG. 13 is a vertical cross-section view of a semiconductor package including a second underfill material portion located between the front side surface of the package substrate and the second side surface of the interposer according to various embodiments of the present disclosure.

(28) FIG. 14 is a vertical cross-section view of a semiconductor package including a semiconductor material interposer according to an alternative embodiment of the present disclosure.

(29) FIG. 15 is a flow diagram illustrating a method of fabricating a semiconductor package according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

(30) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(31) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Unless explicitly stated otherwise, each element having the same reference numeral is presumed to have the same material composition and to have a thickness within a same thickness range.

(32) Various embodiments disclosed herein may be directed to semiconductor devices, and in particular to semiconductor packages and methods of fabricating semiconductor packages having a package substrate that may include one or more recesses in a surface of the package substrate.

(33) Typically, in a semiconductor package a number of semiconductor integrated circuit (IC) dies (i.e., “chips”) may be mounted onto a common substrate, which may also be referred to as a “package substrate.” In some packages, such as in a fan out wafer level package (FOWLP) and/or fan-out panel level package (FOPLP), a plurality of semiconductor IC dies may be mounted to an interposer, such as an organic interposer or a semiconductor (e.g., silicon) interposer, that may include interconnect structures extending therethrough. The resulting package structure, including the interposer and the semiconductor IC dies mounted thereon, may then be mounted onto a surface

of a package substrate using solder connections. An underfill material portion may be provided in the space between the interposer and the package substrate to encapsulate the solder connections and improve the structural coupling between the interposer and the package substrate.

(34) In order to improve electrical performance of the package, some package designs may include additional structures located between the interfacing surfaces of the interposer and the package substrate. For example, one or more semiconductor devices, which may be composed of silicon, may be located over portions of the surface of the interposer that is mounted to the package substrate. In some embodiments, the semiconductor devices may function as a capacitor and may help to improve the signal integrity of the active IC devices of the package. However, the presence of these structures between the interposer and the package substrate may negatively affect the integrity of the structural coupling between the interposer and the package substrate.

(35) In particular, in instances in which a semiconductor device located over the surface of the interposer contacts the package substrate, or in instances when the gap between the semiconductor device and the package substrate is too small to permit adequate flow of the underfill material between the semiconductor device and the package substrate, void areas may be present in the underfill material portion between the interposer and the package substrate. These void areas may lead to package defects, such as delamination of the interposer from the package substrate, cracking of the underfill material portion, and/or moisture-induced “popcorn” defects, which may negatively effect package yields and performance.

(36) Various embodiments disclosed herein include semiconductor packages and methods of fabricating semiconductor packages utilizing a package substrate having one or more recesses formed in a surface of the package substrate and at least one channel in a bottom surface of each of the recesses. Each of the recesses formed in the surface of the package substrate may be configured to accommodate a semiconductor device located over a surface of an interposer that may be bonded to the package substrate. Accordingly, a minimum gap distance may be maintained between the semiconductor device and the package substrate. The minimum gap distance may encourage sufficient underfill material to flow between the semiconductor device and the package substrate and within the at least one channel formed in the bottom surface of each of the recesses. This may improve the integrity of the structural coupling between the interposer and the package substrate, and may significantly reduce the likelihood of package defects, such as delamination, cracking, and/or popcorn defects, thereby improving package yields and performance.

(37) FIG. 1 is a vertical cross-section view of an exemplary intermediate structure during a process of forming a semiconductor package according to various embodiments of the present disclosure. Referring to FIG. 1, the exemplary intermediate structure includes a first carrier substrate **101** and an interposer **103** formed and mounted over a front side surface of the first carrier substrate **101**. The first carrier substrate **101** may provide mechanical support to the interposer **103**, and may be formed of a suitable substrate material, such as glass material, a ceramic material (e.g., a sapphire substrate), a semiconductor material (e.g., a silicon substrate), or the like. Other suitable materials for the first carrier substrate **101** are within the contemplated scope of disclosure. In some embodiments, the first carrier substrate **101** may be formed of an optically transparent material.

(38) In some embodiments, a first release layer **117** may be located over the front side surface of the first carrier substrate **101**, and the interposer **103** may be located over the first release layer **117**. The first release layer **117** may include an adhesive material that may adhere the interposer **103** to the front side surface of the first carrier substrate **101**. In some embodiments, the first release layer **117** may include an adhesive material that may be subsequently treated to cause the adhesive material of the first release layer **117** lose its adhesive properties, such that the first carrier substrate **101** may be separated from the interposer **103**. In some embodiments, the adhesive material of the first release layer **117** may lose its adhesive properties when subjected to treatment using an energy source, such as a thermal, optical (e.g., UV, laser, etc.) and/or sonic (e.g., ultrasonic) energy source. In one non-limiting example, the first release layer **117** may include a light-to-heat conversion

(LTHC) material that may selectively absorb optical radiation in certain wavelength range(s), such as ultraviolet radiation, causing the LTHC material to heat up and thereby lose adhesion. In other embodiments in which the first carrier substrate **101** is formed of an optically transparent material, the application of an optical energy source may cause the first release layer **117** to lose its adhesive property. Alternatively, the first release layer **117** may include an adhesive material, such as an acrylic pressure-sensitive adhesive material, that may decompose when subjected to an elevated temperature. Other suitable materials for the first release layer **117** are within the contemplated scope of disclosure.

(39) Referring again to FIG. **1**, the interposer **103** may include a first side surface **102** and a second side surface **104** opposite the first surface **103**. The second side surface **104** of the interposer **103** may face the front side surface of the first carrier substrate **101**. A plurality of conductive interconnect structures **108** (e.g., metal lines and vias) may extend within the interposer **103** between the first side surface **102** and the second side surface **104** of the interposer **103**. The conductive interconnect structures **108** may be formed in and surrounded by an insulating matrix that may be formed of a dielectric material **118**. The conductive interconnect structures **108** of the interposer **103** may be configured to route electrical signals between semiconductor integrated circuit (IC) dies and a package substrate in a semiconductor package to be subsequently formed. Thus, the conductive interconnect structures **108** of the interposer **103** may also be referred to as “redistribution structures.”

(40) In some embodiments, the interposer **103** may be an organic interposer. The organic interposer **103** may be formed on the first carrier substrate **101**. In one non-limiting example, the interposer **103** may be formed by sequentially depositing layers of a dielectric material **118**, such as a dielectric polymer material, over the front side surface of the first carrier substrate **101** (and over the first release layer **117**, if present). Each of the layers of dielectric material **118** may be lithographically patterned and etched to form open regions (e.g., trenches and/or via openings), and a metallization process may then be used to fill the open regions and form conductive interconnect structures **108** (e.g., metal lines and vias) within each successive layer of dielectric material **118**. In this manner, the interposer **103** may be built layer-by-layer over the front side surface of the first carrier substrate **101**.

(41) In some embodiments, each of the layers of dielectric material **118** of the interposer **103** may include a suitable dielectric polymer material, such as polyimide (PI), benzocyclobutene (BCB), or polybenzobisoxazole (PBO). Other suitable dielectric materials are within the contemplated scope of disclosure. The layers of dielectric material **118** of the interposer **103** may be formed using a suitable deposition process, such as a spin coating and drying process. Other suitable deposition processes are within the contemplated scope of disclosure.

(42) The conductive interconnect structures **108** of the interposer **103** may be formed of a suitable conductive material, such as Cu, Ni, W, Cu, Co, Mo, Ru, etc., including alloys and combinations of the same. In some embodiments, the conductive interconnect structures **108** may include a metallic barrier layer, such as a layer of Ti, TiN, TaN, or WN, contacting the dielectric material **118**, and a metallic fill material, which may include an elemental metal (e.g., Cu, Ni, etc.) or an alloy or a combination thereof. Other suitable materials for the conductive interconnect structures **108** of the interposer **103** are within the contemplated scope of disclosure. The conductive interconnect structures **108** of the interposer **103** may be formed using any suitable deposition process. For example, suitable deposition processes may include physical vapor deposition (PVD), sputtering, chemical vapor deposition (CVD), atomic layer deposition (ALD), plasma-enhanced chemical vapor deposition (PECVD), electrochemical deposition (e.g., electroplating), or combinations thereof.

(43) Referring again to FIG. **1**, an instance of an interposer **103** located over the front side surface of the first carrier substrate **101** may be referred to as a unit area (UA) of the first carrier substrate **101**. A single unit area (UA) is illustrated in FIG. **1**, although it will be understood that the first

carrier substrate **101** may include a plurality of unit areas (UAs), where each unit area (UA) may include a separate instance of an interposer **103** over the front side surface of the first carrier substrate **101**. For example, the first carrier substrate **101** may include a periodic two-dimensional array (such as a rectangular array) of unit areas (UAs), where each unit area (UA) of the array may include a separate instance of an interposer **103** over the front side surface of the carrier substrate **101**. In some embodiments, each interposer **103** within a unit area (UA) of the array may have an identical structure. The plurality of interposers **103** over the first carrier substrate **101** may be continuous with one another, such that a continuous layer of dielectric material **118** may extend over the front side surface of the first carrier substrate **101**, with separate instances of conductive interconnect structures **108** formed within the continuous layer of dielectric material **118** in each unit area (UA).

(44) FIG. 2 is a vertical cross-section view of the exemplary intermediate structure showing interposer bonding structures **106** located over the first side surface **102** of the interposer **103** according to various embodiments of the present disclosure. Referring to FIG. 2, the interposer bonding structures **106** may include a plurality of metallic bumps. The interposer bonding structures **106** may be formed by depositing one or more layers of a metal material and patterning the one or more layers of metal material to form the plurality of interposer bonding structures **106** over the first side surface **102** of the interposer **103**. Each bonding structure **106** may be electrically coupled to an underlying conductive interconnect structure **108** of the interposer **103**. In some embodiments, the interposer bonding structures **106** may form at least one periodic two-dimensional array (such as a rectangular array) of interposer bonding structures **106** within the unit area (UA). In some embodiments, a plurality of interposer bonding structures **106** may be formed over the first side surface **102** of the interposer **103** in each unit area (UA) of the first carrier substrate **101**.

(45) In various embodiments, the interposer bonding structures **106** may be configured for subsequent microbump bonding (i.e., C2 bonding) to corresponding bonding structures formed on semiconductor integrated circuit (IC) dies. In some embodiments, the interposer bonding structures **106** may include a plurality of metal pillars. The metal pillars may include copper or a copper-containing alloy. In some embodiments, the bonding structures may include a plurality of metal stacks, such as a plurality of Cu—Ni—Cu stacks. In some embodiments, the interposer bonding structures **106** may include a solder material, such as tin or a tin-containing alloy, on an upper surface of the interposer bonding structures **106**. Other suitable materials and/or configurations for the interposer bonding structures **106** are within the contemplated scope of disclosure.

(46) FIG. 3 is a vertical cross-section view of the exemplary intermediate structure showing a plurality of semiconductor integrated circuit (IC) dies **105** mounted over the first side surface **102** of the interposer **103** according to various embodiments of the present disclosure. In some embodiments, the plurality of IC semiconductor dies **105** may include at least one system-on-chip (SoC) die. An SoC die may include, for example, an application processor die, a central processing unit die, and/or a graphic processing unit die. In some embodiments, the plurality of IC semiconductor dies **105** may include at least one memory die. The at least one memory die may include a high bandwidth memory (HBM) die. In some embodiments, a HBM die may include a vertical stack of interconnected memory dies. In some embodiments, the plurality of semiconductor IC dies **105** may be homogeneous, meaning that all of the semiconductor IC dies **105** may be of the same type (e.g., all SoC dies, all HBM dies, etc.). Alternatively, the plurality of semiconductor IC dies **105** may be heterogeneous, meaning that the plurality of semiconductor IC dies **105** may include different types of semiconductor IC dies **105** (e.g., at least one SoC die and at least one HBM die). In some embodiments, the plurality of semiconductor IC dies **105** may include one or more SoC dies and a plurality of HBM dies. The one or more SoC dies may be located in a central portion of the unit area (UA) and the plurality of HBM dies may laterally surround the one or more SoC dies. Further, although two semiconductor IC dies **105** are shown mounted over the first side

surface **102** of the interposer **103** in the exemplary embodiment of FIG. 3, it will be understood that in various embodiments more than two semiconductor IC dies **105** may be mounted over the first side surface **102** of the interposer **103**.

(47) Referring again to FIG. 3, each of the semiconductor IC dies **105** may include a plurality of semiconductor die bonding structures **119** located over a lower surface of the semiconductor IC die **105**. The semiconductor die bonding structures **119** on the semiconductor IC dies **105** may have a similar or identical configuration as the interposer bonding structures **106** over the first side surface **102** of the interposer **103** described above with reference to FIG. 2. For example, the semiconductor die bonding structures **119** on the lower surfaces of the semiconductor IC dies **105** may include a plurality of metallic bumps, such as metal pillars and/or metal stacks. In some embodiments, the semiconductor die bonding structures **119** on the semiconductor IC dies **105** may include a solder material, such as tin or a tin-containing alloy, on the lower surface of the semiconductor die bonding structures **119**. The semiconductor die bonding structures **119** on the lower surfaces of each semiconductor IC die **105** may be configured for microbump bonding (i.e., C2 bonding) to corresponding interposer bonding structures **106** on the first side surface **102** of the interposer **103**.

(48) The semiconductor IC dies **105** may be mounted over the first side surface **102** of the interposer **103** by placing each of the semiconductor IC dies **105** over the first side surface **102** of the interposer **103** (e.g., using a pick-and-place apparatus). The semiconductor IC dies **105** may be aligned over the first side surface **102** of the interposer **103** such that the semiconductor die bonding structures **119** on the lower surfaces of the semiconductor IC dies **105** contact corresponding interposer bonding structures **106** over the first side surface **102** of the interposer **103**. A reflow process may be used to bond the semiconductor die bonding structures **119** on the lower surfaces of the semiconductor IC dies **105** to the corresponding interposer bonding structures **106** over the first side surface **102** of the interposer **103**, thereby providing a mechanical and electrical connection between each of the semiconductor IC dies **105** and the interposer **103**. In various embodiments, a plurality of semiconductor IC dies **105** may be mounted over the first side surface **102** of the interposer **103** within each unit area (UA) of the first carrier substrate **101**.

(49) FIG. 4 is a vertical cross-section view of the exemplary intermediate structure showing a first underfill material portion **107** located between the lower surfaces of the semiconductor IC dies **105** and the first side surface **102** of the interposer **103**, and a molding portion **109** around the outer periphery of the plurality of semiconductor IC dies **105** according to various embodiments of the present disclosure. Referring to FIG. 4, the first underfill material portion **107** may be applied into the spaces between the first side surface **102** of the interposer **103** and the plurality of semiconductor IC dies **105** mounted to the interposer **103**. The first underfill material portion **107** may laterally surround and contact each of the interposer bonding structures **106** and semiconductor die bonding structures **119** that bond the respective semiconductor IC dies **105** to the interposer **103**. The first underfill material portion **107** may also be located between adjacent semiconductor IC dies **105** of the plurality of semiconductor IC dies **105** mounted to the interposer **103**.

(50) The first underfill material portion **107** may include any underfill material known in the art. For example, the first underfill material portion **107** may be composed of an epoxy-based material, which may include a composite of resin and filler materials. Other suitable materials for the first underfill material portion **107** are within the contemplated scope of disclosure. Any known underfill material application method may be used to apply the first underfill material portion **107**.

(51) Referring again to FIG. 4, a molding portion **109** may laterally surround the plurality of semiconductor IC dies **105** mounted to the interposer **103**. The molding portion **109** may contact lateral side surfaces of at least some of the semiconductor IC dies **105** and may also contact the first underfill material portion **107**. In various embodiments, the molding portion **109** may include an epoxy material. For example, the molding portion **109** may include an epoxy mold compound

(EMC) that may include epoxy resin, a hardener (i.e., a curing agent), silica or other filler material(s), and optionally additional additives. The EMC may be applied around the periphery of the semiconductor IC dies **105** in liquid or solid form, and may be hardened (i.e., cured) to form a molding portion **109** having sufficient stiffness and mechanical strength surrounding the plurality of semiconductor IC dies **105**. Portions of the molding portion **109** that extend above a horizontal plane including the top surfaces of the semiconductor IC dies **105** may be removed using a planarization process, such as a chemical mechanical planarization (CMP) process.

(52) In various embodiments, each unit area (UA) of the first carrier substrate **101** may include a first underfill material portion **107** located between the first side surface **102** of the interposer **103** and the undersides of the plurality of semiconductor IC dies **105** mounted to the interposer **103**, and a molding portion **109** around the outer periphery of the plurality of semiconductor IC dies **105**. In some embodiments, the molding portion **109** may form a continuous matrix extending between the unit areas (UAs) of the first carrier substrate **101** and laterally surrounding and embedding the respective sets of semiconductor IC dies **105** within each of the unit areas (UAs) of the first carrier substrate **101**.

(53) FIG. 5 is a vertical cross-section view of the exemplary intermediate structure showing a second release layer **121** located over the upper surfaces of the plurality of semiconductor dies **105**, the exposed upper surface of the first underfill material portion **107** and the exposed upper surface of the molding portion **109**, and a second carrier substrate **111** over the second release layer **121** according to various embodiments of the present disclosure. Referring to FIG. 5, the second release layer **121** may include an adhesive material that may adhere the second carrier substrate **111** to the upper surfaces of the plurality of semiconductor dies **105**, the first underfill material portion **107** and the molding portion **109**. As with the first release layer **117** described above, the second release layer **121** may also be configured to lose its adhesive properties when subjected to a treatment using an energy source, such as a thermal, optical (e.g., UV, laser, etc.) and/or sonic (e.g., ultrasonic) energy source. In some embodiments, the first release layer **117** and the second release layer **121** may be composed of the same material(s). Alternatively, the first release layer **117** and the second release layer **121** may be composed of different material(s).

(54) Referring again to FIG. 5, the second carrier substrate **111** may be formed of a suitable substrate material, such as the materials described above with reference to the first carrier substrate **101** shown in FIG. 1. In some embodiments, the second carrier substrate **111** may be composed of the same material(s) as the first carrier substrate **101**. Alternatively, the second carrier substrate **111** and the first carrier substrate **101** may be composed of different material(s). In various embodiments, the second carrier substrate **111** may extend over each of the unit areas (UAs) of the first carrier substrate **101** such that each unit area (UA) of the first carrier substrate **101** may correspond to an equivalent unit area (UA) of the second carrier substrate **111**.

(55) FIG. 6 is a vertical cross-section view of the exemplary intermediate structure showing the first carrier substrate **101** removed according to various embodiments of the present disclosure. Referring to FIG. 6, the first carrier substrate **101** may be removed using any suitable method known in the art. In embodiments in which the first carrier substrate **101** is adhered to the interposer **103** by a first release layer **117**, the first release layer **117** may be subjected to a treatment that causes the first release layer **117** to lose its adhesive properties. This may enable the first carrier substrate **101** to be separated from the exemplary intermediate structure. For example, the first release layer **117** may include a light-to-heat conversion (LTHC) material that may be irradiated by optical radiation in a specified wavelength range, such as ultraviolet radiation, causing the LTHC material to heat up and thereby lose adhesion. The first release layer **117** may optionally be irradiated through the first carrier substrate **101** in embodiments in which the first carrier substrate **101** is composed of an optically-transparent material. Alternatively, the first release layer **117** may include a thermally-decomposing adhesive material. The exemplary intermediate structure may be subjected to a thermal anneal process at a debonding temperature sufficient to cause the first

release layer **117** to decompose and thereby enable the first carrier substrate **101** to be detached from the exemplary intermediate structure. In embodiments in which a thermal anneal process is used to remove the first carrier substrate **101**, the debonding temperature used to thermally decompose the first release layer **117** may not be sufficient cause the second release layer **121** to lose its adhesive properties.

(56) Referring again to FIG. **6**, the exemplary intermediate structure may be inverted (i.e., flipped over), either prior to or following the removal of the first carrier substrate **101**, such that the interposer **103** may be located over and supported by the second carrier substrate **111**.

(57) FIG. **7A** is a vertical cross-section view of the exemplary intermediate structure showing a plurality of bonding pads **115** and semiconductor devices **112** located over the second side surface **104** of the interposer **103** according to various embodiments of the present disclosure. FIG. **7B** is a top view of the exemplary intermediate structure of FIG. **7A**. The vertical cross-section view of the exemplary intermediate structure of FIG. **7A** is taken along line A-A' in FIG. **7B**.

(58) Referring to FIGS. **7A** and **7B**, the bonding pads **115** may be formed by depositing one or more layers of a metal material and patterning the one or more layers of metal material to form the plurality of bonding pads **115** over the second side surface **104** of the interposer **103**. Each bonding pad **115** may be electrically coupled to an underlying conductive interconnect structure **108** of the interposer **103**. The bonding pads **115** may have a rectangular or square horizontal cross-sectional shape as shown in FIGS. **7A** and **7B**. Other suitable horizontal cross-sectional shapes of the bonding pads **115**, such as polygonal, circular, elliptical, and/or irregular shapes, are within the contemplated scope of disclosure. The bonding pads **115** may be formed of a suitable metallic material, such as copper, aluminum, nickel, titanium, etc., including combinations and alloys thereof. Other suitable metallic materials for the bonding pads **115** are within the contemplated scope of disclosure. In some embodiments, the plurality of bonding pads **115** may form a periodic two-dimensional array (such as a rectangular array) of bonding pads **115** within the unit area (UA). The array of bonding pads **115** may be non-contiguous over the unit area (UA). In particular, the array of bonding pads **115** may include one or more gaps corresponding to the location(s) of one or more semiconductor devices **112** located over the second side surface **104** of the interposer **103**. In various embodiments, the array of bonding pads **115** within the unit area (UA) may extend over a greater areal extent within a horizontal plane parallel to the first horizontal direction **hd1** and the second horizontal direction **hd2** than the corresponding areal extent of the array(s) of interposer bonding structures **106** used to mount the plurality of semiconductor IC dies **105** to the interposer **103**. In such embodiments, the bonding pads **115** may also be referred to as “fan-out” bonding pads **115**. In some embodiments, the bonding pads **115** may be configured for subsequent mounting of a unit area (UA) of the exemplary intermediate structure to a package substrate, such via a plurality of C4 (i.e., controlled collapse) solder connections.

(59) Referring again to FIGS. **7A** and **7B**, at least one semiconductor device **112** may be located over the second side surface **104** of the interposer **103**. Each of the semiconductor devices **112** may be mounted to the second side surface **104** of the interposer by a plurality of semiconductor device bonding structures **113**. In some embodiments, the semiconductor device bonding structures **113** may have a similar or identical configuration as the interposer bonding structures **106** located over the first side surface **102** of the interposer **103** described above with reference to FIG. **2**. For example, the semiconductor device bonding structures **113** may include a plurality of metallic bumps, such as metal pillars and/or metal stacks, formed over the second side surface **104** of the interposer **103**. The semiconductor devices **112** may be mounted to respective sets of semiconductor device bonding structures **113** on the second side surface **104** of the interposer **103** via microbump (e.g., C2) bonding connections. Each of the semiconductor device bonding structures **113** may be electrically coupled to an underlying conductive interconnect structure **108** of the interposer **103**. Accordingly, each of the semiconductor devices **112** may be electrically coupled to one or more semiconductor IC dies **105** via the interposer **103**.

(60) In one non-limiting example, the semiconductor devices **112** may be mounted over the second side surface **104** of the interposer **103** by placing each of the semiconductor devices **112** over the first side surface **102** of the interposer **103** such that the lower surfaces of the respective semiconductor devices **112** contact semiconductor device bonding structures **113** located over the second side surface **104** of the interposer **103**. A reflow process may be used to bond the semiconductor devices **112** to respective semiconductor device bonding structures **113** over the second side surface **104** of the interposer **103**, thereby providing a mechanical and electrical connection between each of the semiconductor devices **112** and the interposer **103**.

(61) The at least one semiconductor device **112** may be composed of a suitable semiconductor material. In some embodiments, the at least one semiconductor device **112** may include silicon. Other suitable semiconductor materials are within the contemplated scope of disclosure. In some embodiments, each of the semiconductor devices **112** may have electronic circuit elements, such as conductive interconnect structures, located on and/or within the semiconductor device **112**. Additional electronic circuit elements, such as transistors, capacitors, resistors, diodes, photodiodes, fuse devices, or other similar devices, may also be located on and/or within the semiconductor devices **112** in various embodiments. In some embodiments, the at least one semiconductor device **112** may function as a capacitor and may help to improve signal integrity of the one or more semiconductor IC dies **105** to which the respective semiconductor device **112** is electrically coupled. The at least one semiconductor device **112** mounted on the semiconductor device bonding structures **113** located over the second side surface **104** of the interposer **103** together with the interposer **103** and semiconductor dies **105** (along with other elements as discussed with reference to FIG. **10** below) may form the package structure **150**.

(62) In the embodiment shown in FIGS. **7A** and **7B**, each of the semiconductor devices **112** may have a rectangular horizontal cross-sectional shape. Other suitable horizontal cross-sectional shapes of the semiconductor devices **112**, such as polygonal, circular, elliptical, and/or irregular shapes, are within the contemplated scope of disclosure. Each of the semiconductor devices **112** may have a first maximum length dimension, $L_{sub.S1}$, along a first horizontal direction $hd1$ and a second maximum length dimension, $L_{sub.S2}$, along a second horizontal direction $hd2$ that is orthogonal to the first horizontal direction $hd1$. Each of the semiconductor devices **112** may have an identical size and shape, or may have different (i.e., non-uniform) sizes and/or shapes. For example, in the embodiment shown in FIGS. **7A** and **7B**, the semiconductor devices **112** have non-uniform first and second maximum length dimensions, $L_{sub.S1}$ and $L_{sub.S2}$. In some embodiments, the first and second maximum length dimensions, $L_{sub.S1}$ and $L_{sub.S2}$, of the semiconductor devices **112** may be between about 300 μm and about 5 mm, although greater and lesser maximum length dimensions, $L_{sub.S1}$ and $L_{sub.S2}$, are within the contemplated scope of disclosure. In various embodiments, a height dimension, H , between the upper surfaces of the semiconductor devices **112** and the second side surface **104** of the interposer **103** may be greater than 10 μm , such as between 50 μm and 150 μm , including about 100 μm .

(63) In various embodiments, a plurality of bonding pads **115** may be formed over the second side surface **104** of the interposer **103** and at least one semiconductor device **112** may be mounted over the second side surface **104** of the interposer **103** within each unit area (UA) of the second carrier substrate **111**.

(64) FIG. **8A** is a vertical cross-section view of a package substrate **201** according to various embodiments of the present disclosure. FIG. **8B** is a top view of the package substrate **201** of FIG. **8A**. Referring to FIGS. **8A** and **8B**, the package substrate **201** may include any suitable substrate material(s), such as polymer, glass, epoxy resin, ceramic and/or semiconductor substrate materials. The package substrate **201** may include a first side surface **202** (which, for convenience, may also be referred to as a “front” side surface **202** of the package substrate **201**) and a second side surface **203** (which, for convenience, may also be referred to as a “rear” side surface **203** of the package substrate) that is opposite the first side surface **202**.

(65) The package substrate **201** may be configured such that a plurality of semiconductor IC dies **105** may be mounted over the front side surface **202** of the package substrate **201** to provide a semiconductor package. In various embodiments, the plurality of semiconductor IC dies **105** may be mounted to an interposer **103**, such as an interposer **103** as described above with reference to FIGS. 1-7B, and the interposer **103** having the plurality of semiconductor IC dies **105** mounted thereon may be mounted over the front side surface **202** of the package substrate **201** to provide the semiconductor package. The interposer **103** may be mounted over a bonding region **206** of the package substrate **201**.

(66) In various embodiments, the package substrate **201** may include redistribution structures **204** (e.g., metal lines, vias, bonding regions, etc.) extending within the package substrate **201**. In some embodiments, the rear side surface **203** of the package substrate **201** may be configured to be mounted to a supporting substrate, such as a printed circuit board (PCB). Electrical connections between the supporting substrate (e.g., a PCB) and the semiconductor package may be made via the redistribution structures **204** within the package substrate **201**.

(67) In some embodiments, the package substrate **201** may include a multi-layer structure including a substrate core **213**, at least one redistribution layer **214**, and at least one outer coating layer **215**. For example, the package substrate **201** may include a pair of redistribution layers **214** located above and below the substrate core **213**, and a pair of outer coating layers **215** located above and below the respective redistribution layers **214**. The outer coating layers **215** may form the front side surface **202** and rear side surface **203** of the package substrate **201**. The substrate core **213** may be a plate-like member composed of a suitable material such as an epoxy resin, glass, and/or ceramic material. The substrate core **213** may include a plurality of conductive via structures **216** extending through the substrate core **213**. The redistribution layers **214** may include conductive interconnect structures **217**, such as metal lines, vias and bonding regions, embedded in a dielectric material matrix **218**. In some embodiments, the dielectric material matrix **218** may include multiple layers of a dielectric material, such as a photosensitive epoxy material. Each layer of dielectric material may be lithographically patterned to form open regions (e.g., trenches and via openings) within the respective layers of dielectric material. A metallization process may be used to fill the open regions with a suitable conductive material, such as copper or a copper-alloy, within each layer of dielectric material to form the conductive interconnect structures **217** embedded within the dielectric material matrix **218**. The outer coating layers **215** of the package substrate **201** may include a layer of solder resist material formed over the respective redistribution layers **214**. Each of the layers of solder resist material may provide a protective coating for the package substrate **201** and the underlying redistribution structures **204** within the package substrate **101**. The solder resist material may also inhibit solder material from adhering to the front side surface **202** and rear side surface **203** of the package substrate **201** during a subsequent solder reflow process. An outer coating layer **215** formed of solder resist material may also be referred to as a “solder mask.” Other suitable configurations for the package substrate **201** are within the contemplated scope of disclosure.

(68) In various embodiments, the package substrate **201** may have a total thickness, T , between the front side surface **202** and the rear side surface **203** of the package substrate **201** that is between about 600 μm and about 1,500 μm , although greater and lesser thicknesses for the package substrate **201** are within the contemplated scope of disclosure.

(69) FIG. 9A is a vertical cross-section view of the package substrate **201** showing a plurality of recesses **205** formed in the first side surface **202** of the package substrate **201** according to various embodiments of the present disclosure. FIG. 9B is a top view of the package substrate of FIG. 9A. The vertical cross-section view of the package substrate **201** of FIG. 9A is taken along line B-B' in FIG. 9B.

(70) Referring to FIGS. 9A and 9B, at least one recess **205** may be formed in the first side surface **202** of the package substrate **201**. Each recess **205** may correspond to the location of a semiconductor device **112** over the second side surface **104** of an interposer **103** to be subsequently

mounted over the first side surface **202** of the package substrate **201**. In various embodiments, each recess **205** may have a vertical depth, D , from the first side surface **202** of the package substrate **201** of at least $5\text{ }\mu\text{m}$, which may be sufficient to accommodate the thickness of the semiconductor device **112** as well as the height of the semiconductor bonding structures **113** that bond the semiconductor device **112** to the second side surface **104** of the interposer **103** when the interposer **103** is mounted over the first side surface of the package substrate **201** as described in further detail below. In some embodiments, the depth, D , of each recess **205** may be greater than or equal to $10\text{ }\mu\text{m}$ and less than or equal to $100\text{ }\mu\text{m}$, although greater and lesser depth dimensions, D , of the recesses **205** are within the contemplated scope of disclosure. In some embodiments, a ratio of the depth dimension D of each recess **205** to the total thickness T of the package substrate **201** may be between about and 0.16 , such as between 0.015 and 0.1 . This may provide sufficient clearance to enable an adequate flow of underfill material between the semiconductor device **112** and the bottom surface of the recess **205** without compromising the performance or mechanical integrity of the package substrate **201**.

(71) In various embodiments, the total area of each recess **205** within a horizontal plane parallel to the first horizontal direction $hd1$ and the second horizontal direction $hd2$ may be greater than the total area of the corresponding semiconductor device **112** within a horizontal plane parallel to the first horizontal direction $hd1$ and the second horizontal direction $hd2$. In some embodiments, the total area of the recess **205** may be greater than the total area of the corresponding semiconductor device **112** by at least about 2% . In various embodiments, the maximum length dimensions, $L_{sub.R1}$ and $L_{sub.R2}$, of each recess along the first horizontal direction $hd1$ and the second horizontal direction $hd2$, respectively, may be greater than the maximum length dimensions, $L_{sub.S1}$ and $L_{sub.S2}$, of the corresponding semiconductor device **112** along the first horizontal direction $hd1$ and the second horizontal direction $hd2$, respectively. In some embodiments, the maximum length dimensions, $L_{sub.R1}$ and $L_{sub.R2}$, of the recess **205** may be greater than the maximum length dimensions, $L_{sub.S1}$ and $L_{sub.S2}$, of the corresponding semiconductor device **112** by at least about $50\text{ }\mu\text{m}$. This may enable sufficient underfill material to flow around the sides of the semiconductor devices **112**. In some embodiments, at least one length dimension of the recess **205** may be $350\text{ }\mu\text{m}$ or more.

(72) Each of the recesses **205** may have a suitable horizontal cross-sectional shape, such as a polygonal, circular, elliptical, and/or irregular shape. FIGS. **9A** and **9B** illustrate a pair of recesses **205** having a rectangular horizontal cross-sectional shape. FIG. **9C** is a top view of a package substrate **201** having recesses **205** with an elliptical horizontal cross-section shape according to another embodiment of the present disclosure. FIG. **9D** is a top view of a package substrate **201** having recesses **205** with an irregular horizontal cross-section shape according to yet another embodiment of the present disclosure. Other suitable shapes for the recesses **205** are within the contemplated scope of disclosure. In some embodiments, the horizontal cross-sectional shape of each recess **205** may be the same as the horizontal cross-sectional shape of the corresponding semiconductor device **112**. Alternatively, a recess **205** may have a different horizontal cross-sectional shape than the shape of the corresponding semiconductor device **112**, so long as the recess **205** is large enough to encompass the corresponding semiconductor device **112**. In the embodiment shown in FIGS. **9A** and **9B**, the recesses **205** are shown having vertically-extending sidewalls and sharp, squared corners, although it will be understood that various embodiments may include recesses **205** having tapered or curved sidewalls and/or angled or rounded corners.

(73) The one or more recesses **205** may be formed in the front side surface **202** of the package substrate **201** using any suitable technique. In some embodiments, the one or more recesses **205** may be formed using a lithographic process. For example, a layer of photoresist may be applied over the front side surface **202** of the package substrate **201** and exposed to radiation through an optical mask to transfer the pattern of the one or more recesses **205** to the photoresist layer. The photoresist layer may then be developed to form a patterned mask having openings through the

patterned mask corresponding to the locations of the one or more recesses **205** to be subsequently formed. One or more etching processes may be performed through the patterned mask to selectively remove material from the package substrate **201** to a desired depth, D, thereby forming the one or more recesses **205** on the front side surface **202** of the package substrate **201**. The patterned mask may then be removed using a suitable process, such as via ashing or dissolution using a solvent.

(74) In other embodiments, the one or more recesses **205** may be formed using a machining technique, such as using computer numerical control (CNC) machining of the front side surface **202** of the package substrate **201**. Other known machining techniques, such as drilling or physical etching techniques, may be utilized. In some embodiments, the one or more recesses **205** may be formed using an ablation technique, such as via laser ablation of select portions of the front side surface **202** of the package substrate **201**.

(75) In some embodiments, the one or more recesses **205** may be formed in the front side surface **202** of the package substrate **201** during the process of manufacturing the package substrate **201**. Alternatively, the one or more recesses **205** may be formed after the package substrate **201** is manufactured using post-processing techniques. In embodiments in which the package substrate **201** includes a multilayer structure including a redistribution layer **214** and an outer coating layer **215** (e.g., a solder mask) surrounding a substrate core **213**, the one or more recesses **205** may be formed through the outer coating layer **215** and may optionally extend partially or completely through the redistribution layer **214**. FIG. **10A** is a vertical cross-section view of the package substrate **201** showing a plurality of recesses **205** formed in the first side surface **202** of the package substrate **201** extending partially into the redistribution layer **214** of the package substrate **201** according to an embodiment of the present disclosure. The recesses **205** may be formed by depositing a whole layer of the outer coating layer **215** and then etch the outer coating layer **215**.

(76) In some embodiments, the depth, D, of a recess **205** may extend to and/or into the substrate core **213** of the package substrate **201**. FIG. **10B** is a vertical cross-section view of the package substrate **201** showing a plurality of recesses **205** formed in the first side surface **202** of the package substrate **201** extending partially into the substrate core **213** of the package substrate **201** according to an embodiment of the present disclosure.

(77) In some embodiments, one or more recesses **205** may be formed by depositing an outer coating layer **215** over a surface of a package substrate **201**, and performing an etching process to remove select portions of the outer coating layer **215** to form one or more recesses **205** in the first side surface **202** of the package substrate **201**. The outer coating layer **215** may be a solder resist (i.e., solder mask) material as described above. Other suitable materials for the outer coating layer **215** are within the contemplated scope of disclosure. In various embodiments, the etching process may be performed through a lithographically-patterned mask that may expose the outer coating layer **215** in portions corresponding to the one or more recesses **205** and may protect the remaining portions of the outer coating layer **215** from being etched.

(78) In embodiments in which underlying redistribution structures **204** of the package substrate **201** are exposed on the bottom and/or side surfaces of a recess **205**, such as shown in FIGS. **10A** and **10B**, the redistribution structures may optionally electrically connect to a semiconductor device **112** over the second side surface **104** of an interposer **103** to be subsequently mounted over the first side surface **202** of the package substrate **201**.

(79) FIG. **11A** is a vertical cross-section view of the package substrate **201** showing a channel **260** formed in each recess **205** in the first side surface **202** of the package substrate **201** according to various embodiments of the present disclosure. FIG. **11B** is a top view of the package substrate of FIG. **11A**. The vertical cross-section view of the package substrate **201** of FIG. **11A** is taken along line B-B' in FIG. **11B**. Referring to FIGS. **11A** and **11B**, a channel **260** may be formed within the bottom surface of each of the recesses **205** in the first side surface **202** of the package substrate **201**. Each channel **260** may have an elongated shape including a width dimension, W.sub.C, and a

length dimension, L.sub.C that is greater than the width dimension, W.sub.C. The width dimension W.sub.C of each channel **260** may be less than the corresponding width of the recess **205** in which the channel **260** is located. The length dimension, L.sub.C, of each channel may be less than, equal to or greater than the corresponding length of the recess **205** in which the channel is located. In the embodiment shown in FIG. **11B**, for example, the length dimension L.sub.C of the channel **260** on the left-hand side of FIG. **11B** is less than the corresponding length dimension (i.e., L.sub.R2) of the recess **205** in which the channel **260** is located, whereas the length dimension L.sub.C of the channel **260** on the right-hand side of FIG. **11B** is greater than the corresponding length dimension (i.e., L.sub.R2) of the recess **205** in which the channel **260** is located. In some embodiments, the widths W.sub.C of each channel **260** may be between about 5 μm and about 500 μm , although greater and lesser width dimensions are within the contemplated scope of disclosure. Each channel **260** may have a uniform width W.sub.C along its length, or may have a width dimension that varies along the length of the channel **260**. In some embodiments, the lengths L.sub.C of each channel **260** may be between about 10 μm and about 5 mm, although greater and lesser length dimensions are within the contemplated scope of disclosure. Each channel **260** may follow a straight-line path along its length L.sub.C as shown in FIG. **11B**. Alternatively, one or more channels **260** may follow a curved or angled (e.g., zig-zag) path along its length L.sub.C.

(80) Referring to FIG. **11A**, each of the channels **260** may have a depth dimension, Dc, with respect to the bottom surface of the recess **205** within which the channel **260** is located. In various embodiments, a maximum depth dimension Dc of each channel may be between about 3 μm and about 30 μm , although greater and lesser maximum depth dimensions are within the contemplated scope of disclosure.

(81) The channels **260** may be formed using any of the techniques described above for forming the at least one recess **205**. For example, the channels **260** may be formed via an etching process through a lithographically-patterned mask, via a machining technique, such as CNC machining, milling, drilling, physical etching, etc., or using an ablation technique, such as laser ablation. In some embodiments, the channels **260** may be formed in a separate step following the formation of the one or more recesses **205**. Alternatively, the channels **260** may be formed simultaneous with the formation of the one or more recesses **205**. In other embodiments, the channels **260** may be formed in an initial step, and the one or more recesses **205** may be formed around the channels **260** in subsequent processing steps.

(82) FIGS. **11C-11H** are a tops views of a package substrate **201** illustrating various configurations of recesses **205** and channels **260** as described above with reference to FIGS. **11A** and **11B**. FIG. **11C** is a top view of a package substrate **201** having recesses **205** with an elliptical cross-section shape and a channel **260** located within each recess **205** according to an embodiment of the present disclosure. In the recess **205** on the left-hand side, the channel **260** extends along the second horizontal direction hd2, and in the recess on the right-hand side, the channel **260** extends along the first horizontal direction hd1.

(83) FIG. **11D** is a top view of a package substrate **201** having recesses **205** with an irregular horizontal cross-section shape and a channel **260** located within each recess **205** according to yet another embodiment of the present disclosure. In this embodiment, each of the channels **260** extends at an oblique angle with respect to the first horizontal direction hd1 and the second horizontal direction hd2.

(84) FIG. **11E** is a top view of a package substrate **201** having recesses **205** with multiple channels **260** located within each recess **205** according to yet another embodiment of the present disclosure. Although the embodiment shown in FIG. **11E** includes three channels **260** located within each recess **205**, it will be understood that the recesses **205** may include a greater or lesser number of channels **260**. Further, each of the channels **260** located within a recess **205** may have a uniform size, shape and orientation, or may have non-uniform sizes, shaped and/or orientations.

(85) FIG. **11F** is a top view of a package substrate **201** having recesses **205** with multiple

irregularly-shaped channels **260** located within each recess **205** according to yet another embodiment of the present disclosure.

(86) FIG. **11G** is a top view of a package substrate **201** having recesses **205** with at least one channel **260** extending outside of the boundaries of each recess **205** according to yet another embodiment of the present disclosure. FIG. **11H** is a top view of a package substrate having recesses **205** and at least one channel **260** extending between multiple recesses **205** according to yet another embodiment of the present disclosure.

(87) In various embodiments, the channels **260** formed in the recesses **205** on the first side surface **202** of the package substrate **201** may help to release air and balance the amount of underfill material within the recesses **205** when an interposer **103** including one or more semiconductor devices **112** over the second side surface **104** of an interposer **103** is subsequently mounted over the first side surface **202** of the package substrate **201**.

(88) FIG. **12** is a vertical cross-section view of an exemplary intermediate structure during a process of forming a semiconductor package showing a package structure **150** mounted over the front side surface **202** of a package substrate **201** according to various embodiments of the present disclosure. Referring to FIG. **12**, the second carrier substrate **111** may be removed from the exemplary intermediate package structure **150** shown in FIGS. **7A** and **7B**. The second carrier substrate **111** may be removed using any suitable method known in the art, such as any of the methods described above for removal of the first carrier substrate **101**. In embodiments in which the second carrier substrate **111** is adhered to the semiconductor IC dies **105**, the first underfill material portion **107** and the molding portion **109** using a second release layer **121**, the second release layer **121** may be subjected to a treatment that causes the second release layer **121** to lose its adhesive properties, such as a thermal anneal and/or an optical irradiation treatment process as described above with reference to FIG. **6**.

(89) A dicing process may be used to separate each unit area (UA) of the exemplary intermediate structure to provide a plurality of discrete package structures **150**. Each package structure **150** may include an interposer **103**, a plurality of semiconductor IC dies **105** mounted over a first side surface **102** of the interposer **103**, a first underfill material portion **107** located in the gaps between the first side surface **102** of the interposer **103** and each of the semiconductor IC dies **105**, and a molding portion **109** laterally surrounding the plurality of semiconductor IC dies **105**. The interposer **103** may include a plurality of bonding pads **115** and at least one semiconductor device **112** located over a second side surface **104** of the interposer **103**.

(90) Referring again to FIG. **12**, an etching process may be used to selectively remove portions of the outer coating layer **215** (e.g., solder mask) from the front side surface **202** of the package substrate **201** and expose underlying redistribution structures **204** (e.g., bonding pads). In some embodiments, the same etching process that is used to expose the redistribution structures **204** through the front side surface **202** of the package substrate **201** may also be used to form the at least one recess **205** and/or channel **260** in the package substrate **201** as described above with reference to FIGS. **9A-11H**. Alternatively, the at least one recess **205** and/or channel **260** may be formed in a separate process than the etching process(es) used to expose the redistribution structures **204**. For example, a machining process as discussed above may be used to form the at least one recess **205** and/or channel **260** within the at least one recess **205**. The pattern of the redistribution structures **204** exposed through the front side surface **202** of the package substrate **201** may correspond to the pattern of bonding pads **115** located over the second side surface **104** of the interposer **103**.

(91) Referring again to FIG. **12**, the package structure **150** may be inverted relative to the orientation shown in FIGS. **7A** and **7B** and aligned over the package substrate **201** such that the second side surface **104** of the interposer **103** faces the front side surface **202** of the package substrate **201**. The package structure **150** may be disposed over the front side surface **202** of the package substrate **201** such that an array of solder material portions **207** are located between the

redistribution structures **204** exposed through the front side surface **202** of the package substrate **201** and the bonding pads **115** over the second side surface **104** of the interposer **103**. Each of the semiconductor devices **112** mounted over the second side surface **104** of the interposer **103** may be located over a corresponding recess **205** in the front side surface **202** of the package substrate **201**. Each of the recesses **205** in the front side surface **202** may include at least one channel **260** in a bottom surface of the recess **205**.

(92) A reflow process may be performed to reflow the solder material portions **207**, thereby inducing bonding between the interposer **103** of the package structure **150** and the package substrate **201**. Each of the solder material portions **207** may be bonded to a respective one of the bonding pads **115** over the second side surface **104** of the interposer **103** and to a respective one of redistribution structures **204** (e.g., bonding pads) of the package substrate **201**. In some embodiments, the solder material portions **207** may include C4 solder balls, and the package structure **150** may be bonded to the substrate package **201** through an array of C4 solder balls.

(93) Referring again to FIG. **12**, following the bonding of the package structure **150** to the package substrate **201**, each of the semiconductor devices **112** may be located over a corresponding recess **205** in the front side surface **202** of the semiconductor package **201**. In various embodiments, each of the semiconductor devices **112** may be spaced away from and may not be in contact with the semiconductor package **201**. As shown in FIG. **12**, a gap **209** may be present between the bottom surface of each recess **205** in the package substrate **201** and the lower surface of the overlying semiconductor device **112**. In various embodiments, a height of the gap **209** between the bottom surface of the recess **205** and the lower surface of the overlying semiconductor device **112** may be at least 5 μm . In addition, the peripheral edges of the semiconductor devices **112** may be offset from the sidewalls of the adjacent recess **205** in the package substrate **201** such that no portion of the overlying semiconductor device **112** may extend to or beyond a vertical boundary defined by the peripheral sidewalls of the recess **205**. Each of the semiconductor devices **112** may overlie at least one channel **260** formed in a bottom surface of a recess **205**.

(94) FIG. **13** is a vertical cross-section view of a semiconductor package **100** including a second underfill material portion **211** located between the front side surface **202** of the package substrate **201** and the second side surface **104** of the interposer **103** according to various embodiments of the present disclosure. Referring to FIG. **13**, the second underfill material portion **211** may be applied into the space between the front side surface **202** of the package substrate **201** and the second side surface **104** of the interposer **103**. The second underfill material portion **211** may laterally surround and contact each of the solder material portions **207** that bond the interposer **103** to the package substrate **201** and may also laterally surround and contact each of the semiconductor device bonding structures **113** that bond the respective semiconductor devices **112** to the second side surface **104** of the interposer **103**. The second underfill material portion **211** may flow into each of the recesses **205** in the front side surface **202** of the package substrate **201** and may also flow into the gaps **209** between the bottom surface of each recess **205** in the package substrate **201** and the lower surface of the overlying semiconductor device **112**. The second underfill material portion **211** may also flow into channels **260** in the bottom surface of each recess **205**. Accordingly, each semiconductor device **112** may be surrounded by and contact the second underfill material portion **211** over the upper, lower and lateral side surfaces of the respective semiconductor device **112**. In various embodiments, the portions of the second underfill material portion **211** located within the gaps **209** between the bottom surface of each recess **205** in the package substrate **201** and the lower surface of the overlying semiconductor device **112** may have a thickness of at least 5 μm .

(95) In various embodiments, by providing recesses **205** in the front side surface **202** of the package substrate **201** corresponding to the locations of semiconductor devices **112**, a minimum gap distance may be maintained between the semiconductor devices **112** and the package substrate **201**. This may ensure that a sufficient amount of the second underfill material portion **211** may flow between each semiconductor device **112** and the front side surface **202** of the package

substrate **201**. The one or more channels **260** in the bottom surfaces of each of the recesses **205** may help to release air and balance the amount of the second underfill material portion **211** within the recesses **205**. This may improve the integrity of the structural coupling between the interposer **103** and the package substrate **201** in the semiconductor package **100**, and may significantly reduce the likelihood of package defects, such as delamination, cracking, and/or popcorn defects, thereby improving package yields and performance.

(96) The second underfill material portion **211** may include any underfill material known in the art. For example, the second underfill material portion **211** may be composed of an epoxy-based material, which may include a composite of resin and filler materials. Other suitable materials for the second underfill material portion **211** are within the contemplated scope of disclosure. Any known underfill material application method may be used to apply the second underfill material portion **211**.

(97) In some embodiments, electrical connections to the semiconductor package **100** may be made by mounting the rear side surface **203** of the semiconductor package **200** onto a support substrate containing electrical interconnects, such as a printed circuit board (PCB).

(98) FIG. **14** is a vertical cross-section view of a semiconductor package **200** including a semiconductor material interposer **230** according to an alternative embodiment of the present disclosure. Referring to FIG. **14**, the semiconductor package **200** according to the alternative embodiment may be similar to the semiconductor package **100** shown in FIG. **13**, and may include a package structure **150** having interposer **230**, a plurality of semiconductor IC dies **105** mounted over a first side surface **102** of the interposer **230**, a first underfill material portion **107** located in the gaps between the first side surface **102** of the interposer **230** and each of the semiconductor IC dies **105**, and a molding portion **109** laterally surrounding the plurality of semiconductor IC dies **105**. The interposer **230** may include a plurality of bonding pads **115** and at least one semiconductor device **112** located over a second side surface **104** of the interposer **230**. The package structure **150** may be bonded to the front side surface **202** of a package substrate **201** via a plurality of solder material portions **207**. The front side surface **202** of the package substrate **201** may include at least one recess **205**, where each recess **205** in the package substrate **201** corresponds to the location of a semiconductor device **112** mounted to the second side surface **104** of the interposer **230**. Each of the recesses **205** in the front side surface **202** may include at least one channel **260** in a bottom surface of the recess **205**.

(99) A second underfill material portion **211** may be located in the space between the front side surface **202** of the package substrate **201** and the second side surface **104** of the interposer **103**, including within the gap(s) **209** between the bottom surface of each recess **205** and the lower surface of the overlying semiconductor device **112**, and within each of the channels **260** in a bottom surface of the respective recesses **205**.

(100) The semiconductor package **200** of FIG. **14** may differ from the semiconductor package **100** of FIG. **13** in that the semiconductor package **200** of FIG. **14** includes a semiconductor material interposer **230** as opposed to an organic interposer **103**. Referring to FIG. **14**, the semiconductor material interposer **230** may include a semiconductor device **231**, such as a silicon member, having a plurality of conductive through-vias **233** (e.g., through-silicon vias) extending therethrough. The conductive through-vias **233** may carry electrical signals between the plurality of semiconductor IC dies **105** mounted to the first side surface **102** of the interposer **230** and the package substrate **201**. In some embodiments, the semiconductor material interposer **230** may include at least one redistribution layer **235** including interconnect structures embedded in a dielectric material matrix located above and/or below the semiconductor device **231**.

(101) FIG. **15** is a flowchart illustrating a method **300** of fabricating a semiconductor package **100**, **200** according to various embodiments of the present disclosure. Referring to FIGS. **1-3** and **15**, in step **301** of embodiment method **300**, at least one semiconductor integrated circuit (IC) die **105** may be mounted over a first side surface **102** of an interposer **103**, **230**. Referring to FIGS. **7A**, **7B**

and 15, in step 303 of embodiment method 300, at least one semiconductor device 112 may be mounted over a second side surface 104 of the interposer 103, 230. Referring to FIGS. 9A-10B and 15, in step 305 of embodiment method 300, at least one recess 205 may be formed in a front side surface 202 of a package substrate 201. Referring to FIGS. 11A-11H and 15, in step 307 of embodiment method 300, at least one channel 260 may be formed in a bottom surface of a recess 205 in the front side surface 202 of the package substrate 201. Referring to FIGS. 12 and 15, in step 309 of embodiment method 300, the second side surface 104 of the interposer 103, 230 may be bonded to the front side surface 202 of the package substrate 201 such that each semiconductor device 112 overlies a recess 205 in the front side surface 202 of the package substrate 201.

Referring to FIGS. 13-15 in step 311 of embodiment method 300, an underfill material portion 211 may be provided between the front side surface 202 of the package substrate 201 and the second side surface 104 of the interposer 103, 230 such that the underfill material portion 211 is located between a lower surface of each semiconductor device 112 and a bottom surface of a recess 205 in the front side surface 202 of the package substrate 201 underlying the respective semiconductor device 112, and the underfill material portion 211 is located in the at least one channel 260.

(102) Referring to all drawings and according to various embodiments of the present disclosure, a semiconductor package 100, 200 may include an interposer 103, 230, at least one semiconductor integrated circuit (IC) die 105 mounted over a first side surface 102 of the interposer 103, 230, a semiconductor device 112 mounted over a second side surface 104 of the interposer 103, 230 that is opposite the first side surface 102 of the interposer 103, 230, and a package substrate 201 including a recess 205 in a front side surface 202 of the package substrate 201 and at least one channel 260 in a bottom surface of the recess 205, the second side surface 104 of the interposer 103, 230 bonded to the front side surface 202 of the package substrate 201 such that the semiconductor device 112 overlies the recess 205 and the at least one channel 260 in the front side surface 202 of the package substrate 201.

(103) In an embodiment, the semiconductor package 100, 200 includes an underfill material portion 211 between the front side surface 202 of the package substrate 201 and the second side surface 104 of the interposer 103, 230 and within a gap 209 between a lower surface of the semiconductor device 112 and the bottom surface of the recess 205 in the front side surface 202 of the package substrate 201 and within the at least one channel 260 in the bottom surface of the recess 205. In another embodiment, a height of the gap 209 between the lower surface of the semiconductor device 112 and the bottom surface of the recess 205 in the front side surface 202 of the package substrate 201 is at least 5 μm . In another embodiment, the semiconductor device 112 is spaced from the package substrate 201. In another embodiment, a depth of the recess 205 from the front side surface 202 of the package substrate 201 is at least 5 μm . In another embodiment, the depth of the recess 205 from the front side surface 202 of the package substrate 201 is equal to or greater than 10 μm and less than or equal to 100 μm . In another embodiment, a ratio of the depth of the recess 205 to a total thickness of the package substrate 201 is between 0.0067 and 0.16. In another embodiment, the semiconductor device 112 has a length dimension along a first horizontal direction that is at least 300 μm , and the recess 205 has a length dimension along the first horizontal direction that is at least 50 μm greater than the length dimension of the semiconductor device 112 along the first horizontal direction. In another embodiment, a plurality of semiconductor IC dies 105 are bonded to the first side surface 102 of the interposer 103, 230 via semiconductor die bonding structures 119, a plurality of semiconductor devices 112 are bonded to the second side surface 104 of the interposer via second semiconductor device bonding structures 113, each of the plurality of semiconductor devices 112 including at least one circuit element located on or in the semiconductor device 112, and each of the plurality of semiconductor devices 112 is electrically coupled to at least one semiconductor IC die via interconnect structures extending through the interposer 103, 230, the package substrate 201 includes a plurality of recesses 205 in the front side surface 202 of the package substrate 201, each recess 205 of the plurality of recesses 205 including

at least one channel **260** in a bottom surface of the recess **205**, where each semiconductor device **112** of the plurality of semiconductor devices **112** bonded to the second side surface **104** of the interposer **103, 230** overlies a respective recess **205** and at least one channel **260** on the front side surface **202** of the package substrate **201**, the package substrate comprises a plurality of solder material portions **207** located between respective bonding pads **115** of the second side surface **104** of the interposer **103, 230** and interconnect structures **204** of the package substrate **201**, and the underfill material portion **211** laterally surrounds and contacts the solder material portions **207** and the semiconductor device bonding structures **113** and is located within each gap **209** between a lower surface of a semiconductor device **112** of the plurality of semiconductor devices **112** and a bottom surface of a recess **205** of the plurality of recesses **205** in the front side surface **202** of the package substrate **201**. In another embodiment, the plurality of semiconductor IC dies **105** includes at least one of a system-on-chip (SoC) die and a high bandwidth memory (HBM) die, and wherein at least one semiconductor device **112** of the plurality of semiconductor devices **112** functions as a capacitor to improve signal integrity of the one or more semiconductor IC dies **105**. In another embodiment, the at least one channel **260** has a width dimension WC that is less than a corresponding width dimension of the recess **205** in the front side surface **202** of the package substrate **201**, has a length dimension LC that is less than, equal to, or greater than a corresponding length dimension of the recess **205** in the front side surface **202** of the package substrate **201**, and has a maximum depth dimension DC with respect to the bottom surface of the recess **205** in the front side surface **202** of the package substrate **201** that is at least 3 μm . In another embodiment, the at least one channel **260** extends outside of the boundaries of the recess **205** in the front side surface **202** of the package substrate **201**.

(104) An additional embodiment is drawn to a substrate **201** for a semiconductor package including a front side surface **202**, a rear side surface **203**, a plurality of redistribution structures **204** located between the front side surface **202** and the rear side surface **203** of the substrate **201**, at least one recess **205** in the front side surface **202** of the substrate **201**, the at least one recess **205** having a depth of at least 5 μm and a length dimension along at least one horizontal direction of at least 350 μm , and at least one channel **260** in a recess **205** in the front side surface **202** of the substrate **201**, wherein a long axis direction of the at least one channel **260** is parallel with the front side surface **202** of the substrate **201** in a cross-section view. In an embodiment, the at least one channel **260** extends outside of the boundaries of the recess **205** in the front side surface **202** of the package substrate **201** in a top view. In an embodiment, a total thickness of the substrate **201** between the front side surface **202** and the rear side surface **203** is between 600 μm and 1,500 μm , and a ratio of a depth of the at least one recess **205** to a total thickness of the substrate **201** is between 0.0067 and 0.16. In another embodiment, the substrate **201** includes a substrate core **213**, a redistribution layer **214** over the substrate core **213**, and an outer coating layer **215** over the redistribution layer **214** that forms the front side surface **202** of the substrate **201**, and the at least one recess **205** extends at least through the outer coating layer **215** and into the redistribution layer **214**.

(105) An additional embodiment is drawn to a method of fabricating a semiconductor package **100, 200** that includes mounting at least one semiconductor integrated circuit (IC) die **105** over a first side surface **102** of an interposer **103, 230**, mounting at least one semiconductor device **112** over a second side surface **104** of the interposer **103, 230** that is opposite the first side surface **102**, forming at least one recess **205** in a front side surface of a package substrate **201**, forming at least one channel **260** in a bottom surface of each recess **205** of the at least one recess **205** in the front side surface **202** of the package substrate **201**, bonding the second side surface **104** of the interposer **103, 230** to the front side surface **202** of the package substrate **201** such that each semiconductor device **112** overlies a recess **205** of the at least one recess **205** in the front side surface **202** of the package substrate **201**, and providing an underfill material portion **211** between the front side surface **202** of the package substrate **201** and the second side surface **104** of the interposer **103, 230** such that the underfill material portion **211** is located between a lower surface

of each semiconductor device **112** and a bottom surface of a recess **205** of the at least one recess **205** in the front side surface of the package substrate underlying the respective semiconductor device **112**, and the underfill material portion **211** is located in the at least one channel **260**. In an embodiment, the method further includes forming the interposer **103, 230** over a first carrier substrate **101**, forming a plurality of interposer bonding structures **106** over the first side surface **102** of the interposer **103, 230**, wherein a plurality of semiconductor IC dies **105** are mounted over the first side surface **102** of the interposer **103, 230** via the plurality of first interposer bonding structures **106**, providing a first underfill material portion **107** between the first side surface **102** of the interposer **103, 230** and the underside surfaces of the plurality of semiconductor IC dies **105** and between the respective semiconductor IC dies **105**, and forming a molding portion **109** laterally surrounding the plurality of semiconductor IC dies **105**.

(106) In another embodiment, the method further includes providing a second carrier substrate **111** over upper surfaces of the plurality of semiconductor IC dies **105**, the first underfill material portion **107** and the molding portion **109**, removing the first carrier substrate **101** from the second side surface **104** of the interposer **103, 230**, forming semiconductor device bonding structures **113** over the second side surface **104** of the interposer **103, 230**, where the at least one semiconductor device **112** is mounted over the second side surface **104** of the interposer **103, 230** via the semiconductor device bonding structures **113**, forming a plurality of bonding pads **115** over the second side surface **104** of the interposer **103, 230**, where the second side surface **104** of the interposer **103, 230** is bonded to the front side surface **202** of the package substrate **201** via a plurality of solder material portions **207** located between respective bonding pads **115** over the second side surface **104** of the interposer **103, 230** and redistribution structures **204** exposed through the front side surface **202** of the package substrate **201**, and removing the second carrier substrate **111** from over the upper surfaces of the plurality of semiconductor IC dies **105**, the first underfill material portion **107** and the molding portion **109**.

(107) In another embodiment, forming the at least one recess **205** in the front side surface **202** of the package substrate includes forming each recess **205** of the at least one recess **205** to have a greater horizontal cross-sectional area than a horizontal cross-section area of a semiconductor device **112** overlying the respective recess **205** of the at least recess **205** upon bonding of the second side surface **104** of the interposer **103, 230** to the front side surface **202** of the package substrate **201**, and where forming the at least one channel **260** in a bottom surface of each recess of the at least one recess **205** in the front side surface **202** of the package substrate **201** includes forming the at least one channel **260** having a width dimension WC that is less than a corresponding width dimension of the recess **205** in which the at least one channel **260** is formed, a length dimension LC that is less than, equal to, or greater than a corresponding length dimension of the recess **205** in which the at least one channel **260** is formed, and a maximum depth dimension DC with respect to the bottom surface **205** of the recess **205** in which the at least one channel **260** is formed that is at least 3 μm .

(108) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor package, comprising: an interposer; at least one semiconductor integrated circuit (IC) die mounted over a first side surface of the interposer; a semiconductor device mounted over a second side surface of the interposer that is opposite the first side surface of the interposer; and a package substrate comprising a recess in a front side surface of the package substrate and at least one channel in a bottom surface of the recess, the second side surface of the interposer bonded to the front side surface of the package substrate such that the semiconductor device overlies the recess and the at least one channel in the front side surface of the package substrate and the semiconductor device is spaced from the at least one channel.
2. The semiconductor package of claim 1, further comprising: an underfill material portion between the front side surface of the package substrate and the second side surface of the interposer and within a gap between a lower surface of the semiconductor device and the bottom surface of the recess in the front side surface of the package substrate, and within the at least one channel and within the recess.
3. The semiconductor package of claim 1, wherein a height of the gap between the lower surface of the semiconductor device and the bottom surface of the recess is at least 5 μm .
4. The semiconductor package of claim 1, wherein the semiconductor device is spaced from the package substrate.
5. The semiconductor package of claim 1, wherein a depth of the recess from the front side surface of the package substrate to the bottom surface of the recess is at least 5 μm .
6. The semiconductor package of claim 5, wherein the depth of the recess from the front side surface of the package substrate is equal to or greater than 10 μm and less than or equal to 100 μm .
7. The semiconductor package of claim 1, wherein a ratio of a depth of the recess to a total thickness of the package substrate is between 0.0067 and 0.16.
8. The semiconductor package of claim 1, wherein the semiconductor device has a length dimension along a first horizontal direction that is at least 300 μm , and the recess has a length dimension along the first horizontal direction that is at least 50 μm greater than the length dimension of the semiconductor device along the first horizontal direction.
9. The semiconductor package of claim 2, wherein: a plurality of semiconductor IC dies are bonded to the first side surface of the interposer via semiconductor die bonding structures; a plurality of semiconductor devices are bonded to the second side surface of the interposer via semiconductor device bonding structures, each of the plurality of semiconductor devices comprising at least one circuit element located on or in the semiconductor device, and each of the plurality of semiconductor devices is electrically coupled to at least one semiconductor IC die via interconnect structures extending through the interposer; the package substrate comprises a plurality of recesses in the front side surface of the package substrate, each recess of the plurality of recesses including at least one channel in a bottom surface of the recess, wherein each semiconductor device of the plurality of semiconductor devices bonded to the second side surface of the interposer overlies a respective recess and at least one channel on the front side surface of the package substrate; the package substrate comprises a plurality of solder material portions located between respective bonding pads of the second side surface of the interposer and interconnect structures of the package substrate; and the underfill material portion laterally surrounds and contacts the solder material portions and the semiconductor device bonding structures and is located within each gap between a lower surface of a semiconductor device of the plurality of semiconductor devices and a bottom surface of a recess of the plurality of recesses in the front side surface of the package substrate.
10. The semiconductor package of claim 9, wherein the plurality of semiconductor IC dies comprises at least one of a system-on-chip (SoC) die and a high bandwidth memory (HBM) die, and wherein at least one semiconductor device of the plurality of semiconductor devices comprises a capacitor to improve signal integrity of the one or more semiconductor IC dies.
11. The semiconductor package of claim 1, wherein the at least one channel has a width dimension

- that is less than a corresponding width dimension of the recess in the front side surface of the package substrate, and has a length dimension that is less than, equal to, or greater than a corresponding length dimension of the recess in the front side surface of the package substrate.
12. The semiconductor package of claim 1, wherein the at least one channel extends outside of the boundaries of the recess in the front side surface of the package substrate.
13. A substrate for a semiconductor package, comprising: a front side surface; a rear side surface; a plurality of redistribution structures located between the front side surface and the rear side surface of the substrate; at least one recess in the front side surface of the substrate, the at least one recess having a depth of at least 5 μm and a length dimension along at least one horizontal direction of at least 350 μm ; and at least one channel in the at least one recess in the front side surface of the substrate, wherein a long axis direction of the at least one channel is parallel with the front side surface of the substrate in a cross-section view.
14. The substrate of claim 13, wherein the at least one channel extends outside of the boundaries of the recess in the front side surface of the package substrate in a top view.
15. The substrate of claim 13, wherein a total thickness of the substrate between the front side surface and the rear side surface is between 600 μm and 1,500 μm , and a ratio of a depth of the at least one recess to a total thickness of the substrate is between 0.0067 and 0.16.
16. The substrate of claim 13, wherein the substrate comprises a substrate core, a redistribution layer over the substrate core, and an outer coating layer over the redistribution layer that forms the front side surface of the substrate, and the at least one recess extends at least through the outer coating layer and into the redistribution layer.
17. The semiconductor package of claim 1, wherein the at least one channel has a maximum depth dimension with respect to the bottom surface of the recess in the front side surface of the package substrate that is at least 3 μm .
18. The semiconductor package of claim 1, wherein the at least one channel has a width dimension that is at least 5 μm .
19. The substrate of claim 13, wherein the at least one channel has a maximum depth dimension with respect to the bottom surface of the recess in the front side surface of the package substrate that is at least 3 μm .
20. A semiconductor package, comprising: an interposer; at least one semiconductor integrated circuit (IC) die mounted over a first side surface of the interposer; a semiconductor device mounted over a second side surface of the interposer that is opposite the first side surface of the interposer; and a package substrate, the second side surface of the interposer mounted over a front side of the package substrate, the package substrate comprising a recess in the front side of the package substrate having a greater horizontal cross-sectional area than a horizontal cross-section area of the semiconductor device, the recess comprises at least one channel in a bottom surface of the recess, and a width dimension of the semiconductor device along a first horizontal direction is greater than a width dimension of the channel along the first horizontal direction.
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